

DARREN L. OLDAG

17702 Quitman Mountain Way
Round Rock, Texas 78664
(512) 989-9030
darren at oldag dot net

SUMMARY

Experienced and motivated software developer with proven track record of reliable and maintainable solutions. Excels at architecting robust and extensible systems with a keen understanding of the given problem but with an eye for future enhancement.

TECHNICAL SKILLS

<i>Languages</i>	C++/C, Java, Perl, Assembly
<i>Platforms</i>	Linux, Win32, Solaris, MacOS X, Netware
<i>Architectures</i>	x86, PowerPC, Sparc
<i>Other</i>	XML, GNU Make

EXPERIENCE

Simdesk, Austin, Texas

Senior Software Engineer, August 2005 to Present

- Designed and implemented enhancements to the Simdesk World Wide Server™ Personal Information Management (PIM) component.
 - Designed and implemented email distribution list storage and synchronization.
 - Designed attendee list management for meeting support.
 - Designed extensible properties for any PIM object.
- Designed a calendaring object model conformat to the RFC 2445 iCalendar specification.

Newisys, a Sanmina-SCI Company, Austin, Texas

Design Verification Engineer, September 2004 to July 2005

- Ownership, including design and implementation (Java), of testbench and testcases for simulation and verification of a hardware (Verilog) memory controller module used in a custom ASIC.

Pervasive Software, Austin, Texas

Senior Software Engineer, March 2000 to August 2004

- Ownership and development of Pervasive Services, a platform abstraction layer providing a common API and class library for development on Linux, Win32, Solaris, Netware, and MacOS X. Library includes functionality for abstracting processor architecture, file handling, IPC, process control, threading, synchronization, etc.
- Maintenance and extension of cross platform component build system based on GNU Make. Extensions include customizable frameworks, seamless external and cross-tree consumption, and Java build integration (both native and Ant).
- Designed and implemented a pluggable and skinnable test framework, thus providing a common infrastructure for unit and functional tests. Common framework allowed for automatable, repeatable results, reducing the necessity for quality assurance and developer interaction. Implemented a console UI and XML script engine for running tests. Also implemented a client/server-based UI for remote and distributed testing.
- Architected a product-wide, cross-platform internationalization solution, based on dynamically-loaded XML string resources. This reduced translation work, since all resources were now in a common format, and also allowed Pervasive.SQL to utilize Japanese text on Linux in the supporting utilities for the first time in the history of the product.
- Implemented character codeset conversion routines for various European locales, Japanese

Shift-JIS and Extended Unix Code (EUC-JP), and the various Unicode transformation formats (UTF8, UTF16, and UTF32).

- Authored an NLM import library generation utility, allowing the export of a binary C++ classes from Netware Loadable Modules. This removed the necessity of manually maintaining a text file containing the list of exports in C++ mangled names.
- Integrated Xerces XML SAX parser into the class library for seamless type matching, character conversion, and parser callback authoring.
- Designed an implemented a remote method invocation mechanism for our cross-platform COM implementation. Architected for protocol and transport pluggability. Provided scalable thread model, allowing natural transition from in-process to cross-process execution without behavioral changes.
- Other services implemented for class library include Blowfish encryption algorithm, MD5 secure hash algorithm, transport/channel network abstractions, and a compression library.
- Developed an SDK for Pervasive Services, including an installer, documentation, wizards, etc. Installer eased setup of developer build environment and configuration, including automatic tool detection and integration.
- Conducted performance analysis at both component and system levels, providing insight and solutions to solve performance problems, especially concerning SMP scalability.
- Acted as mentor for C++ and object oriented software design methodologies. Purveyor of best practices.

IBM - RS/6000 Server Development, Austin, Texas

System Engineer, June 1999 to March 2000

- Bring-up, debug, and test support of RS/6000 servers. Required skills include knowledge and analysis of computer system architecture, bus architectures, and hardware/firmware interactions
- Responsible for first power on of new systems, from service processor boot sequence, to main processor configuration and startup, and finally to main operating system hand-off. Debugged and fixed each level until next stage could be successfully reached.
- Component and interconnect debugging on mother boards, processor cards, and add-on cards.
- Used logic analyzers and oscilloscopes to locate design and manufacturing defects. Implemented fixes using available extraneous logic.
- Utilized kernel debugger to set register values on various components to put them into a debugging state.

EDUCATION

Texas A&M University, College Station, Texas

Computer Engineering, B.S., May 1999, 3.896 GPR

Relevant Coursework:

Operating Systems

Data Structures

Software Engineering

Algorithm Analysis

Artificial Intelligence

Networks and Distributed Processing

Computer Architecture

Digital Systems Design

Microelectronic Circuits

Microprocessor Systems Design

Senior design project was a stand-alone Ethernet device capable of sampling data and sending it across a standard IP network. The project was implemented in Verilog and synthesized to a Xilinx FPGA. Each hardware module was simulated using Java in order to validate the design as well as ease the transition to actual hardware/Verilog implementation.